

1. Найдите все записи из лога загрузки, доступного через команду `journalctl` с опцией `-b` в первые 20 секунд с момента загрузки, начало загрузки определить автоматически.

```
root@eltex-practice2-pg1-v16:/# journalctl -b
Oct 15 07:52:01 localhost kernel: Linux version 6.8.0-55-generic (buildd@lcy02-and64->
Oct 15 07:52:01 localhost kernel: Command line: BOOT_IMAGE=/vmlinuz-6.8.0-55-generic >
Oct 15 07:52:01 localhost kernel: KERNEL supported cpus:
Oct 15 07:52:01 localhost kernel: Intel GenuineIntel
Oct 15 07:52:01 localhost kernel: AMD AuthenticAMD
Oct 15 07:52:01 localhost kernel: Hygon HygonGenuine
Oct 15 07:52:01 localhost kernel: Centaur CentaurHauls
Oct 15 07:52:01 localhost kernel: zhaoxin Shanghai
Oct 15 07:52:01 localhost kernel: BIOS-provided physical RAM map:
Oct 15 07:52:01 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x00000000000009fb>
Oct 15 07:52:01 localhost kernel: BIOS-e820: [mem 0x00000000000009fc00-0x00000000000009ff>
Oct 15 07:52:01 localhost kernel: BIOS-e820: [mem 0x0000000000000f0000-0x0000000000000fff>
Oct 15 07:52:01 localhost kernel: BIOS-e820: [mem 0x0000000000100000-0x0000000000bffd9f>
Oct 15 07:52:01 localhost kernel: BIOS-e820: [mem 0x000000000bffd9a000-0x000000000bffffff>
Oct 15 07:52:01 localhost kernel: BIOS-e820: [mem 0x000000000feffc0000-0x000000000fefffff>
Oct 15 07:52:01 localhost kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffff>
Oct 15 07:52:01 localhost kernel: BIOS-e820: [mem 0x0000000100000000-0x000000013fffffff>
Oct 15 07:52:01 localhost kernel: BIOS-e820: [mem 0x0000000fd0000000-0x0000000fffffff>
Oct 15 07:52:01 localhost kernel: NX (Execute Disable) protection: active
Oct 15 07:52:01 localhost kernel: APIC: Static calls initialized
Oct 15 07:52:01 localhost kernel: SMBIOS 2.8 present.
Oct 15 07:52:01 localhost kernel: DMI: QEMU Standard PC (i440FX + PIIX, 1996), BIOS r>
Oct 15 07:52:01 localhost kernel: Hypervisor detected: KVM
Oct 15 07:52:01 localhost kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Oct 15 07:52:01 localhost kernel: kvm-clock: using sched offset of 659413027316 cycles
Oct 15 07:52:01 localhost kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff map>
Oct 15 07:52:01 localhost kernel: tsc: Detected 3399.996 MHz processor
Oct 15 07:52:01 localhost kernel: e820: update [mem 0x00000000-0x00000fff] usable ==>>
Oct 15 07:52:01 localhost kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
Oct 15 07:52:01 localhost kernel: AGP: No AGP bridge found
Oct 15 07:52:01 localhost kernel: last_pfn = 0x140000 max_arch_pfn = 0x400000000
Oct 15 07:52:01 localhost kernel: MTRR map: 4 entries (3 fixed + 1 variable; max 19),>
Oct 15 07:52:01 localhost kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB W>
Oct 15 07:52:01 localhost kernel: last_pfn = 0xbffda max_arch_pfn = 0x400000000
Oct 15 07:52:01 localhost kernel: found SMP MP-table at [mem 0x000f5450-0x000f545f]
Oct 15 07:52:01 localhost kernel: RAMDISK: [mem 0x2fccd000-0x33e5dfff]
Oct 15 07:52:01 localhost kernel: ACPI: Early table checksum verification disabled
Oct 15 07:52:01 localhost kernel: ACPI: RSDP 0x000000000000f5250 000014 (v00 BOCHS )
Oct 15 07:52:01 localhost kernel: ACPI: RSDT 0x0000000000ffe3c25 000038 (v01 BOCHS BX>
Oct 15 07:52:01 localhost kernel: ACPI: FACP 0x0000000000ffe3a07 000074 (v01 BOCHS BX>
Oct 15 07:52:01 localhost kernel: ACPI: DSDT 0x0000000000ffdf040 0049c7 (v01 BOCHS BX>
Oct 15 07:52:01 localhost kernel: ACPI: FACS 0x0000000000ffdf000 000040
Oct 15 07:52:01 localhost kernel: ACPI: APIC 0x0000000000ffe3a7b 000080 (v03 BOCHS BX>
Oct 15 07:52:01 localhost kernel: ACPI: SSDT 0x0000000000ffe3afb 0000ca (v01 BOCHS VM>
Oct 15 07:52:01 localhost kernel: ACPI: HPET 0x0000000000ffe3bc5 000038 (v01 BOCHS BX>
Oct 15 07:52:01 localhost kernel: ACPI: WAET 0x0000000000ffe3bfd 000028 (v01 BOCHS BX>
Oct 15 07:52:01 localhost kernel: ACPI: Reserving FACP table memory at [mem 0xbfffe3a0>
Oct 15 07:52:01 localhost kernel: ACPI: Reserving DSDT table memory at [mem 0xbffdf04>
Oct 15 07:52:01 localhost kernel: ACPI: Reserving FACS table memory at [mem 0xbffdf00>
```

2. Используя `awk` найдите все источники и их сообщения в файле `auth.log` (найдите его `find`), в названии источника удалите информацию об идентификаторе процесса при помощи `sed`, полученный результат отсортируйте по названию источника





3. Для результата из предыдущего пункта найдите количество повторений для каждого источника и выведите их в виде списка «число\_повторений источник», результат отсортируйте по убыванию количества повторений

```
root@eltex-practice2-pg1-v16:/# awk '{ print $2 }' /var/log/auth.log | sort -r |
: uniq -c
      287 eltex-practice2-pg1-v16
root@eltex-practice2-pg1-v16:/#
```

4. В файле /etc/passwd найдите всех пользователей в системе, у которых установлен shell /usr/sbin/nologin и выведите их в виде списка: «UID, username, список его групп», отсортированный в обратном порядке по UID, список групп используйте из файла /etc/group

```
root@eltex-practice2-pg1-v16:/# awk 'BEGIN {FS=":"} {print $3, $1, $5}' /etc/passwd | sed '#/usr/sbin/nologin#' | sort
-r
998 systemd-network systemd Network Management
997 systemd-timesync systemd Time Synchronization
992 systemd-resolve systemd Resolver
991 polkitd User for polkitd
989 fwupd-refresh Firmware update daemon
9 news news
8 mail mail
7 lp lp
65534 nobody nobody
6 man man
5 games games
42 _apt
4 sync sync
39 irc ircd
38 list Mailing List Manager
34 backup backup
33 www-data www-data
3 sys sys
2 bin bin
13 proxy proxy
109 sshd
108 usbmux usbmux daemon,,,
107 landscape
106 tss TPM software stack,,,
105 tcpdump
104 uidd
103 syslog
102 pollinate
101 messagebus
1003 user3 user3,user3,1,1,1
1002 user2 user2,user2,1,1,1
1001 user1
1000 tech tech
100 dhcpcd DHCP Client Daemon,,,
10 uucp uucp
1 daemon daemon
0 root root
root@eltex-practice2-pg1-v16:/#
```

5. Найдите в результате вывода dmesg все строки, содержащие слово 'kernel'

```

12.pyc.139853087203376" pid=41857 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[237881.447351] audit: type=1400 audit(1760752601.529:432): apparmor="DENIED" operation="mknod" class="file"
" profile="ubuntu_pro_apt_news" name="/usr/lib/python3/dist-packages/uclient/_pycache_/event_logger.cpyth
hon-312.pyc.135573685246000" pid=41856 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[237881.448456] audit: type=1400 audit(1760752601.530:433): apparmor="DENIED" operation="mknod" class="file"
" profile="ubuntu_pro_esm_cache" name="/usr/lib/python3/dist-packages/uclient/_pycache_/exceptions.cpyth
on-312.pyc.139853087632944" pid=41857 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[237881.448762] audit: type=1400 audit(1760752601.531:434): apparmor="DENIED" operation="mknod" class="file"
" profile="ubuntu_pro_apt_news" name="/usr/lib/python3/dist-packages/uclient/_pycache_/yaml.cpython-312.
pyc.135573674592432" pid=41856 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[269993.280443] workqueue: update_balloon_size_func hogged CPU for >10000us 512 times, consider switching t
o WQ_UNBOUND
[335773.818800] kauditd_printk_skb: 89 callbacks suppressed
[335773.818803] audit: type=1400 audit(1760850495.609:524): apparmor="DENIED" operation="mknod" class="file"
" profile="ubuntu_pro_apt_news" name="/usr/lib/python3/dist-packages/uclient/_pycache_/__init__.cpython-
312.pyc.139740727916336" pid=44509 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[335773.822114] audit: type=1400 audit(1760850495.614:525): apparmor="DENIED" operation="mknod" class="file"
" profile="ubuntu_pro_apt_news" name="/usr/lib/python3/dist-packages/uclient/_pycache_/apt.cpython-312.p
yc.139740728090096" pid=44509 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[335773.837037] audit: type=1400 audit(1760850495.629:526): apparmor="DENIED" operation="mknod" class="file"
" profile="ubuntu_pro_esm_cache" name="/usr/lib/python3/dist-packages/uclient/_pycache_/__init__.cpython
-312.pyc.127011352104368" pid=44511 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[335773.838331] audit: type=1400 audit(1760850495.630:527): apparmor="DENIED" operation="mknod" class="file"
" profile="ubuntu_pro_esm_cache" name="/usr/lib/python3/dist-packages/uclient/_pycache_/log.cpython-312.
pyc.127011353655680" pid=44511 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[335773.845243] audit: type=1400 audit(1760850495.637:528): apparmor="DENIED" operation="mknod" class="file"
" profile="ubuntu_pro_esm_cache" name="/usr/lib/python3/dist-packages/uclient/_pycache_/defaults.cpython
-312.pyc.127011350992944" pid=44511 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[335773.845562] audit: type=1400 audit(1760850495.637:529): apparmor="DENIED" operation="mknod" class="file"
" profile="ubuntu_pro_esm_cache" name="/usr/lib/python3/dist-packages/uclient/_pycache_/secret_manager.c
python-312.pyc.127011350992944" pid=44511 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[335773.850773] audit: type=1400 audit(1760850495.642:530): apparmor="DENIED" operation="mknod" class="file"
" profile="ubuntu_pro_esm_cache" name="/usr/lib/python3/dist-packages/uclient/_pycache_/system.cpython-3
12.pyc.127011350994992" pid=44511 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[335773.853445] audit: type=1400 audit(1760850495.645:531): apparmor="DENIED" operation="mknod" class="file"
" profile="ubuntu_pro_apt_news" name="/usr/lib/python3/dist-packages/uclient/_pycache_/event_logger.cpyth
on-312.pyc.139740726270000" pid=44509 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[335773.854761] audit: type=1400 audit(1760850495.646:532): apparmor="DENIED" operation="mknod" class="file"
" profile="ubuntu_pro_apt_news" name="/usr/lib/python3/dist-packages/uclient/_pycache_/yaml.cpython-312.
pyc.139740715894960" pid=44509 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[335773.856489] audit: type=1400 audit(1760850495.648:533): apparmor="DENIED" operation="mknod" class="file"
" profile="ubuntu_pro_apt_news" name="/usr/lib/python3/dist-packages/uclient/_pycache_/util.cpython-312.
pyc.139740715894960" pid=44509 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
root@eltex-practice2-pg1-v16:/#

```

## 6. Подсчитайте количество строк в файле /var/log/kern.log

```

root@eltex-practice2-pg1-v16:/# wc -l /var/log/kern.log
11 /var/log/kern.log
root@eltex-practice2-pg1-v16:/#

```

## 7. Отформатируйте вывод записей в /var/log/apt/history.log в следующем порядке, построчно: Commandline: ... ; Start-Date: ... ; End-Date: ...

```

root@eltex-practice2-pg1-v16:/# awk '/^Commandline:/{print "Commandline: " $0}; /^StartDate:/{print "StartDate: " $0}; /^EndDate:/{print "EndDate: " $0}' /var/log/apt/history.log
root@eltex-practice2-pg1-v16:/# cat /var/log/apt/history.log | head -n 20

Start-Date: 2025-02-16 20:51:22
Commandline: apt-get install --reinstall apt
Reinstall: apt:amd64 (2.7.14build2)
End-Date: 2025-02-16 20:51:24

Start-Date: 2025-02-16 20:51:26
Commandline: apt-get install --reinstall base-files
Reinstall: base-files:amd64 (13ubuntu10)
End-Date: 2025-02-16 20:51:28

Start-Date: 2025-02-16 20:51:31
Commandline: apt-get install --reinstall base-passwd
Reinstall: base-passwd:amd64 (3.6.3build1)
End-Date: 2025-02-16 20:51:32

Start-Date: 2025-02-16 20:51:35
Commandline: apt-get install --reinstall bash
Reinstall: bash:amd64 (5.2.21-2ubuntu4)
End-Date: 2025-02-16 20:51:37
root@eltex-practice2-pg1-v16:/#

```